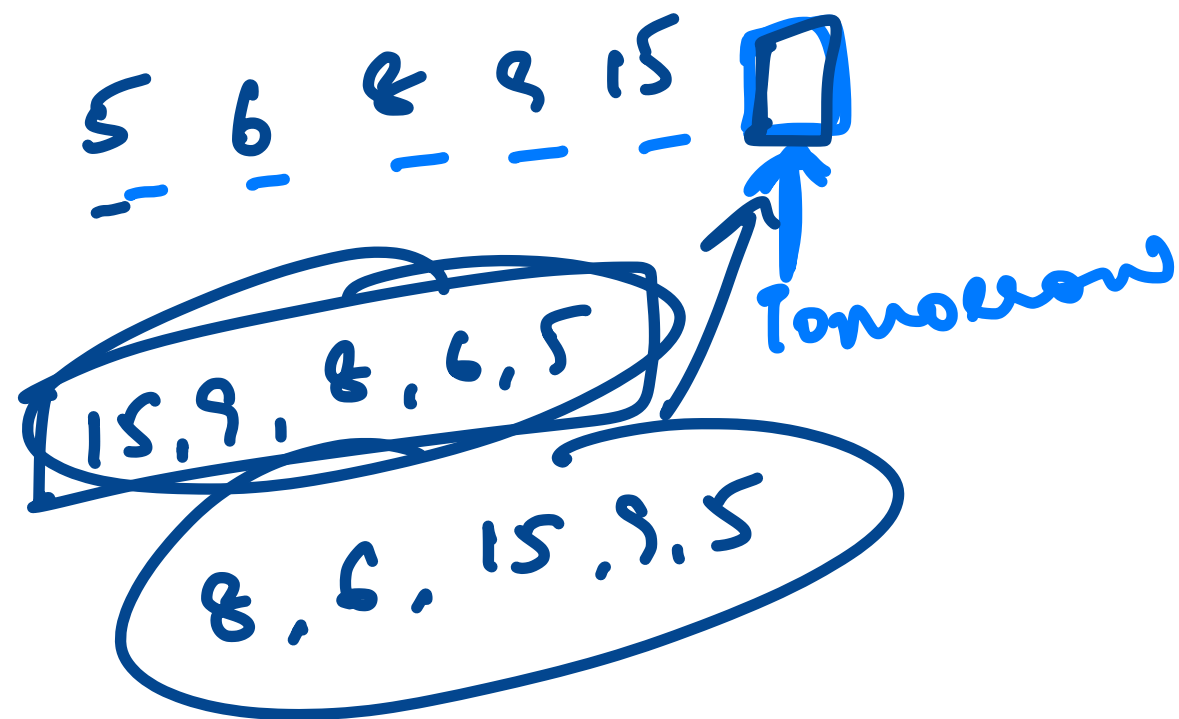
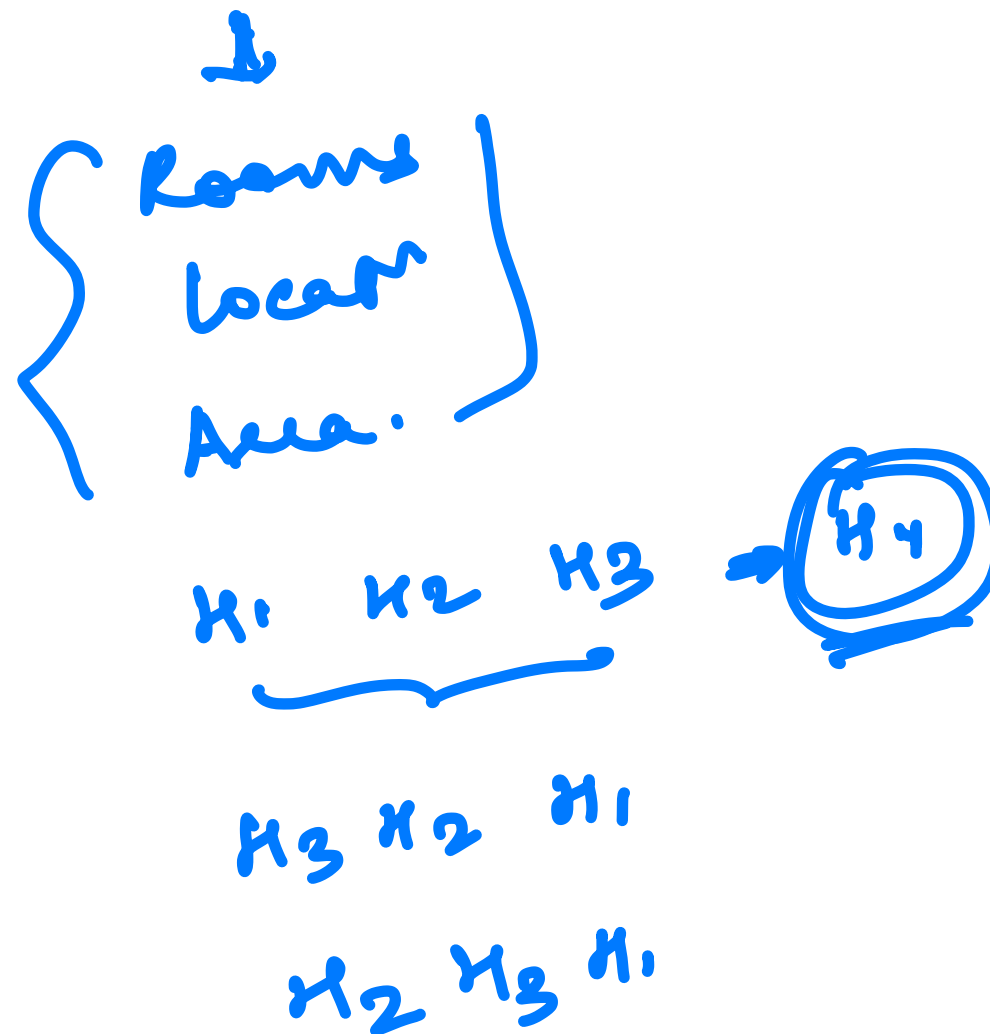


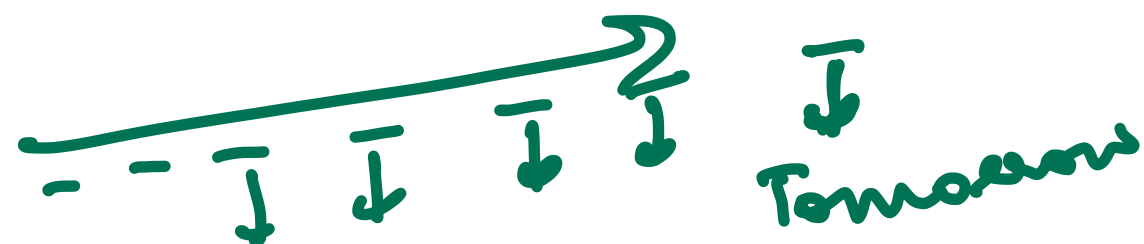
Stock price



House Price Prediction



Weather →



Order / Sequence

Google Translate

/ Autocomplete

English → Hindi

I like ~~Pasta~~
~~Pizza~~

my fav cuisine is

↑
indian italian

DNA Sequence

Music

videos

Alexa / Siri / Google Assistant

Speech / translation

Sequence / Order

ANN / CNN

① I love pizza

② I love food

③ I eat pizza

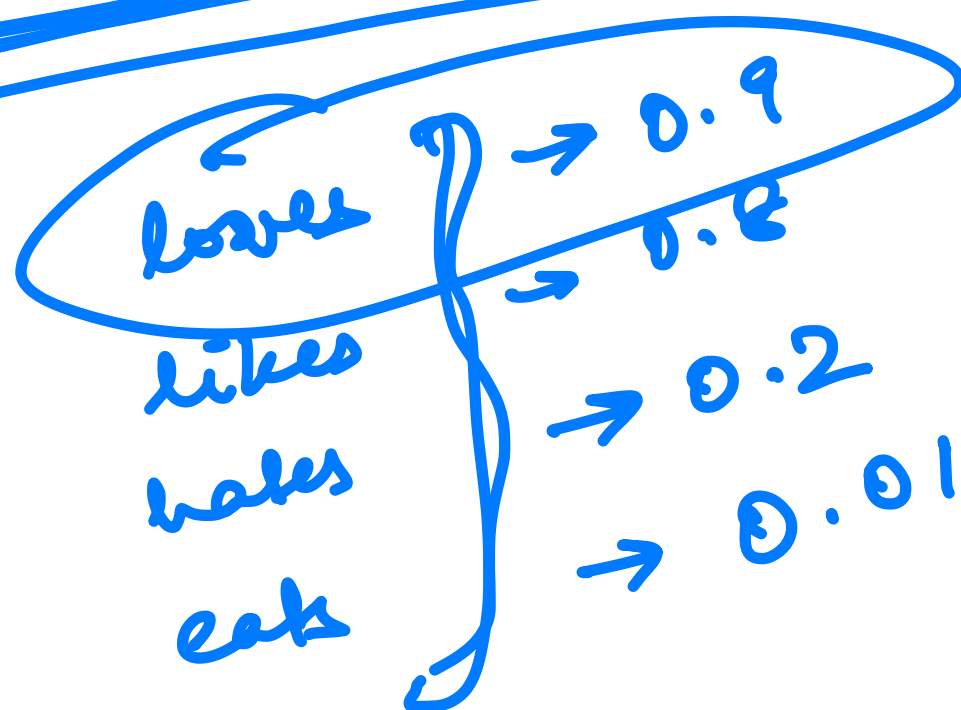
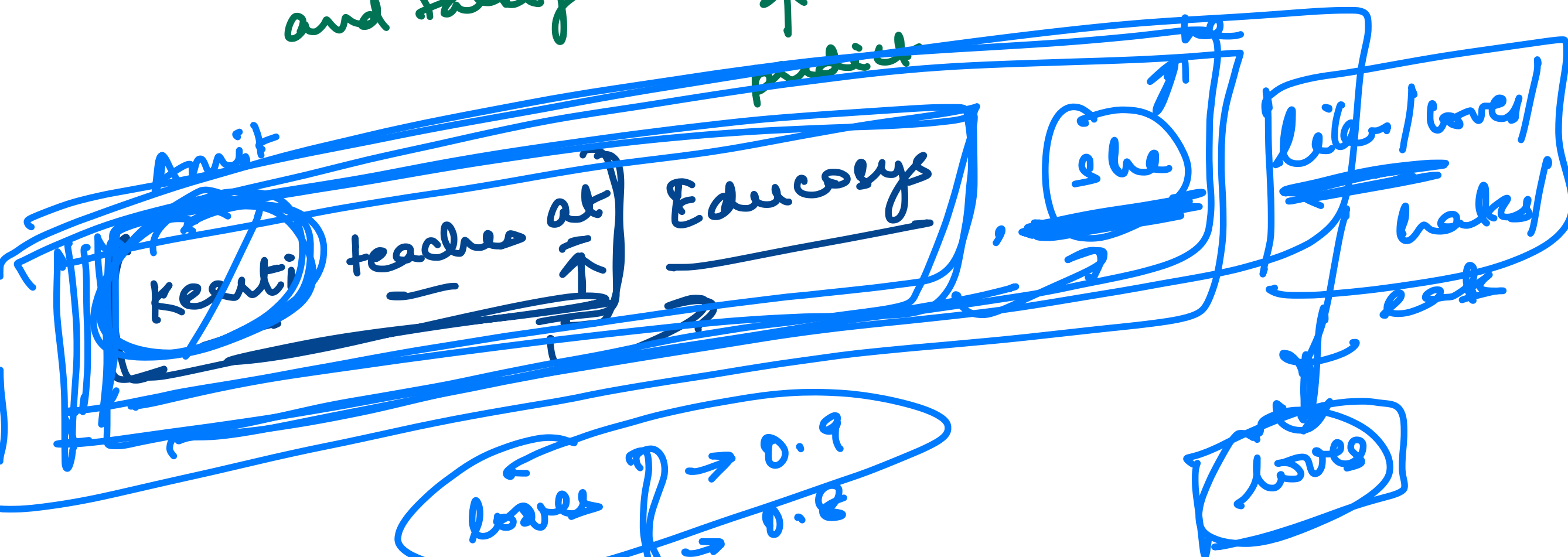
④ I eat food

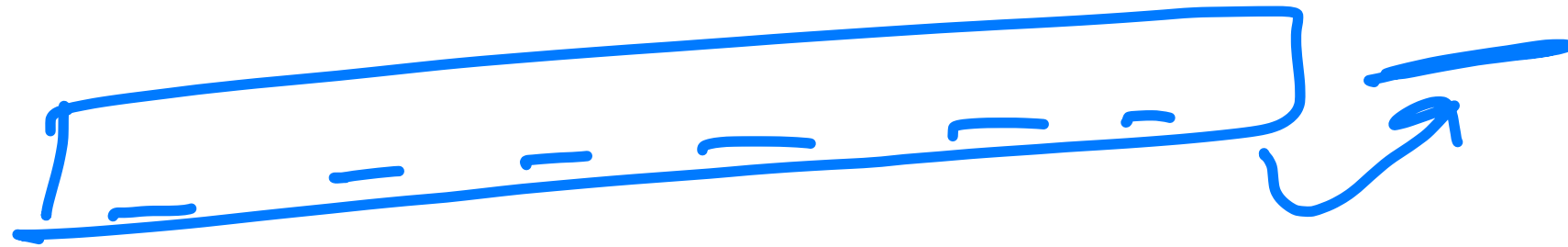
① content from pair words
② order also matters



Reunion

Keerti teaches at Educorys, she likes teaching
and talking to public



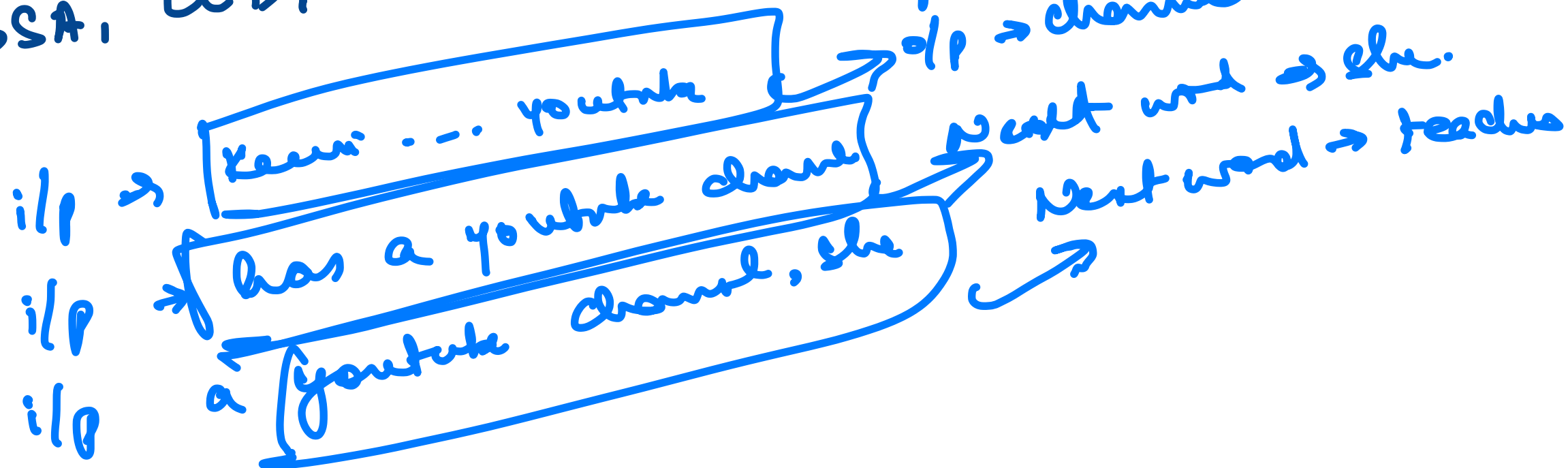


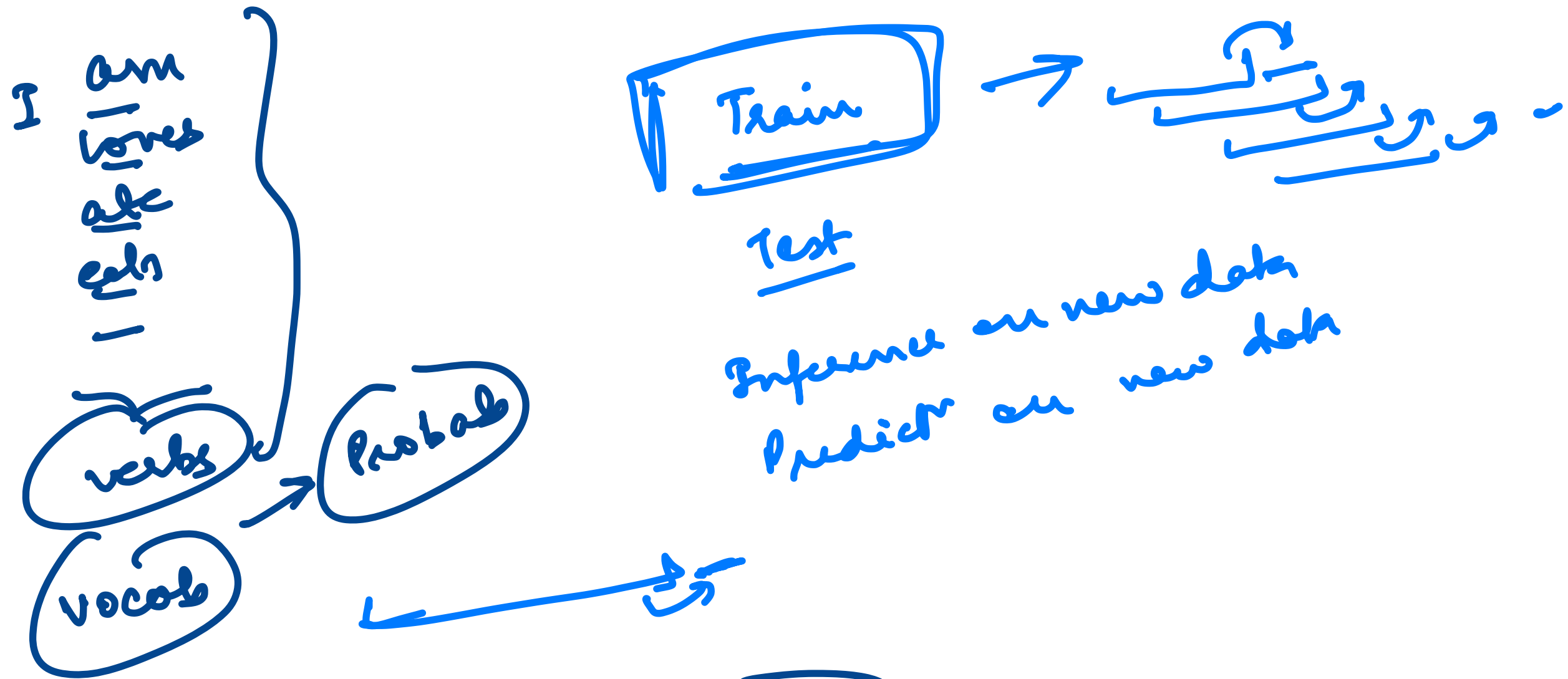
Inference

Training

Keerti has a youtube channel. she teaches

DSA, LLD, NCD





String: (Text) Book

→ Numbers

vector → I/P

- ① Tokenization
- ② Encoding

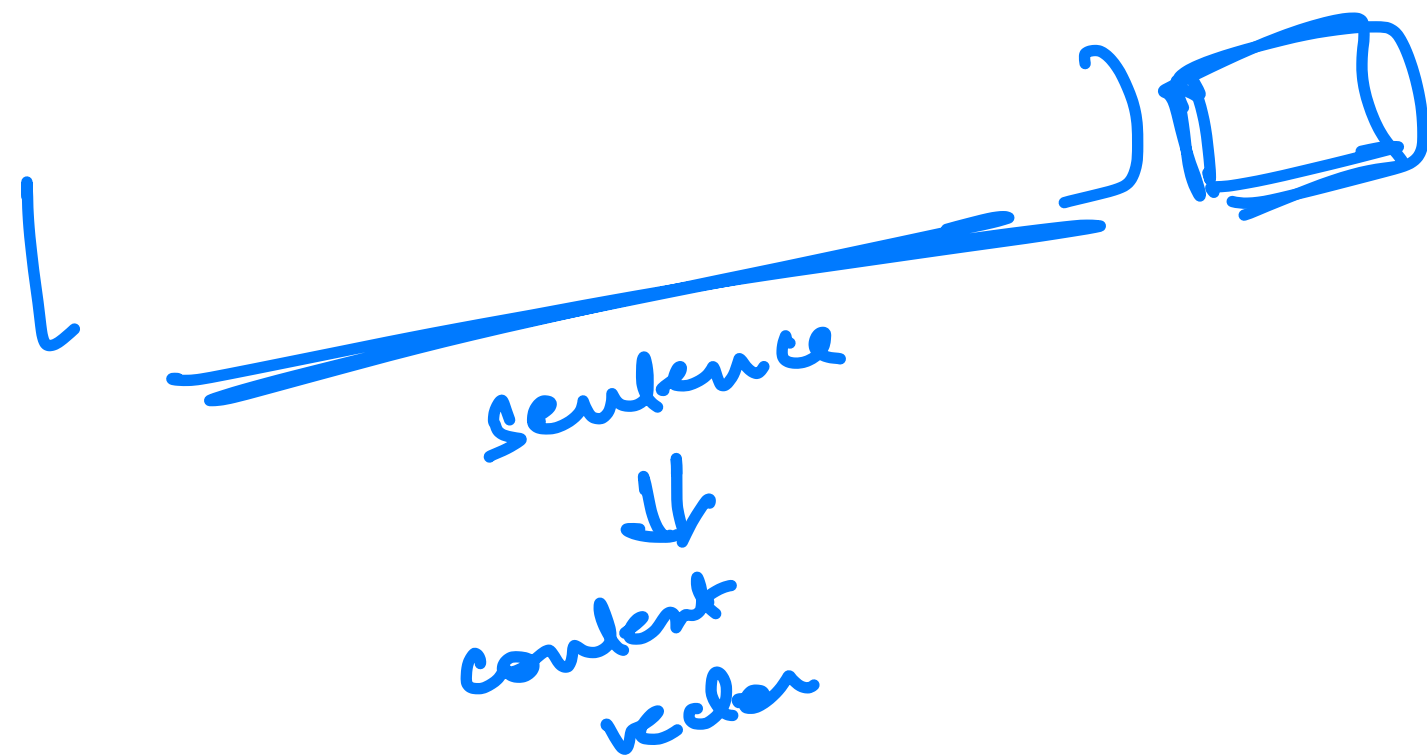
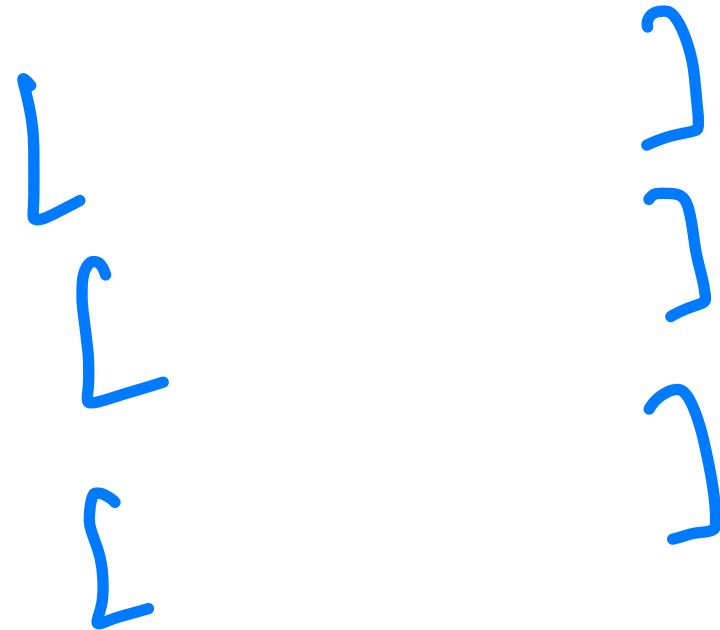
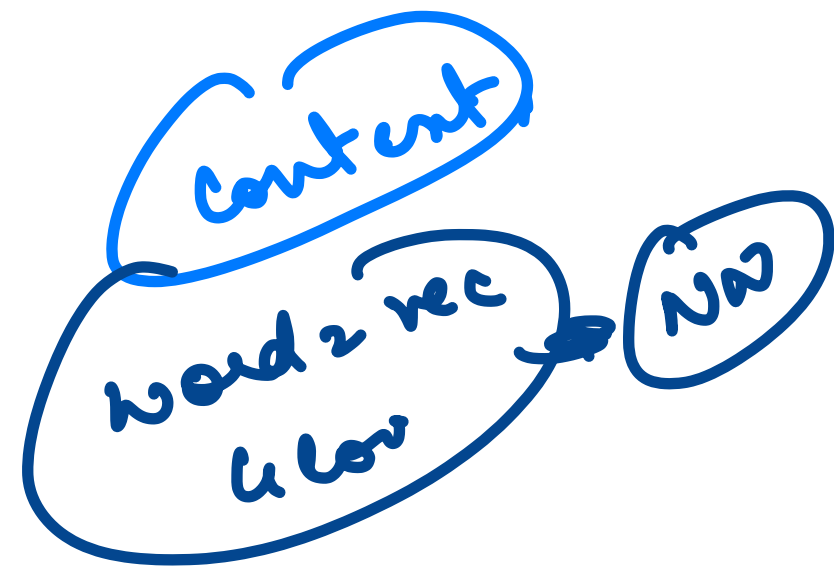
I / love / reaching

↓ ↓ ↓

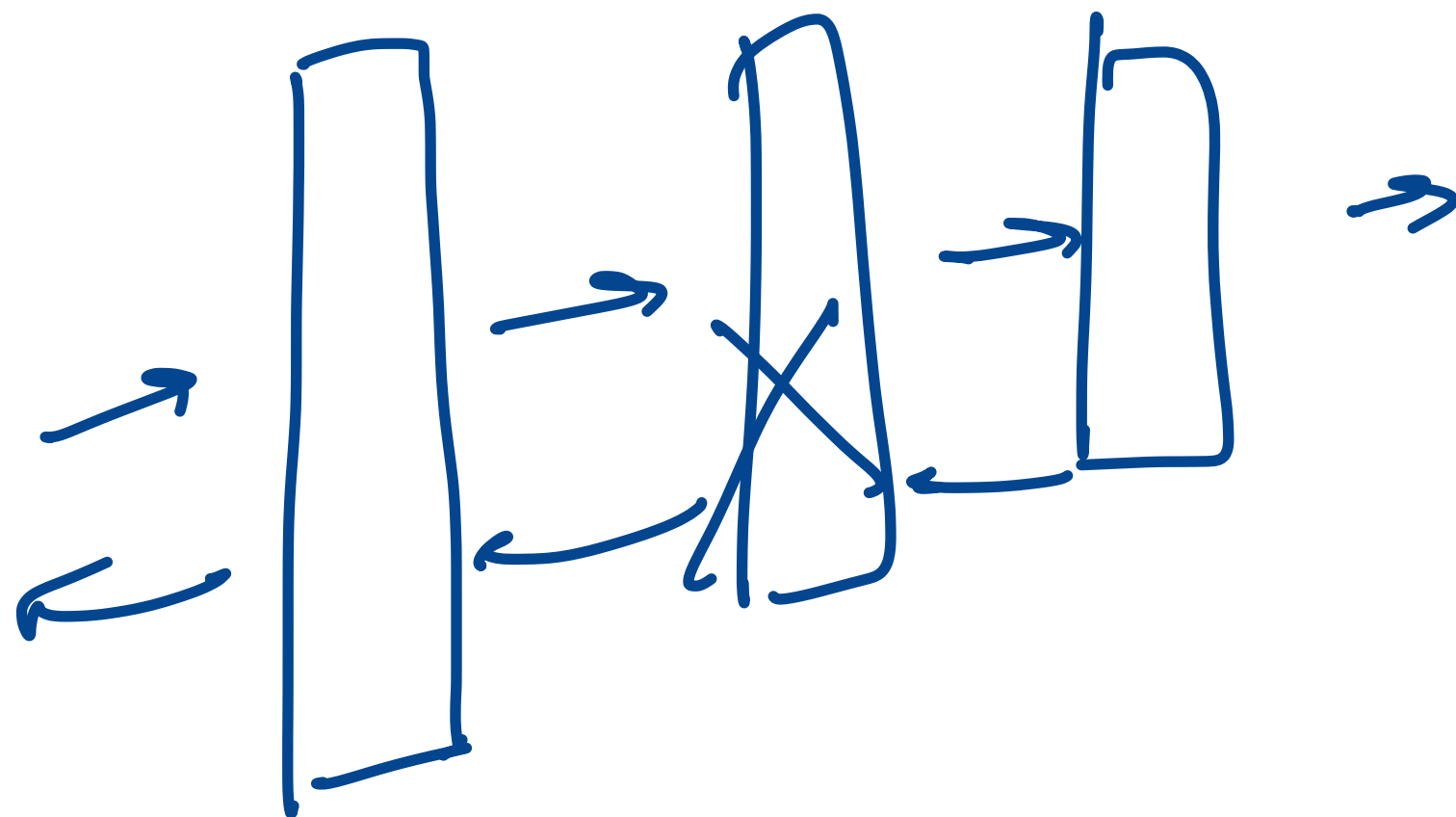
[1, 0, 0] [0, 1, 0] [0, 0, 1]

one hot encoding

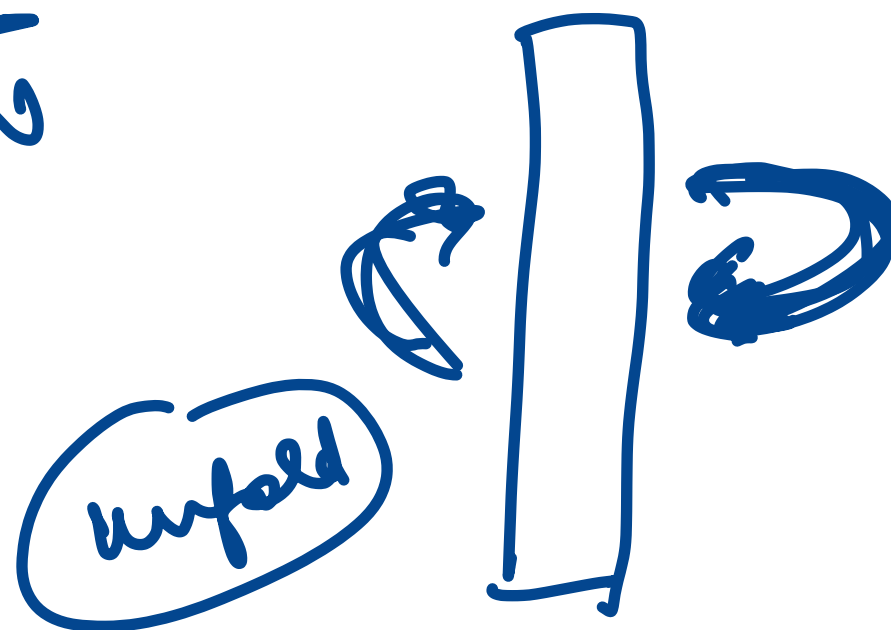
③ Embedding →



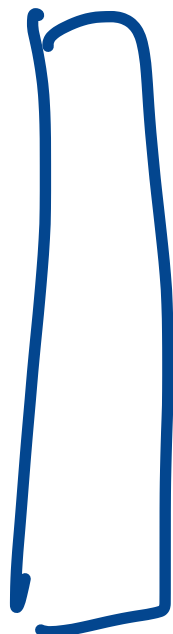
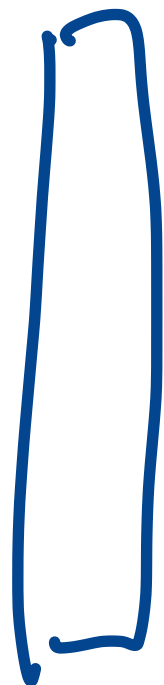
RNN $\Rightarrow$ Recurrent Neural Nlne



Layer
RNN



one layer



Steps



↓
I/p

Content

Hidden State

↓
Content



x_{t-2}

x_{t-1}

x_t

x_{t+1}

x_{t+2}

x_{t+3}

...

word no. 5

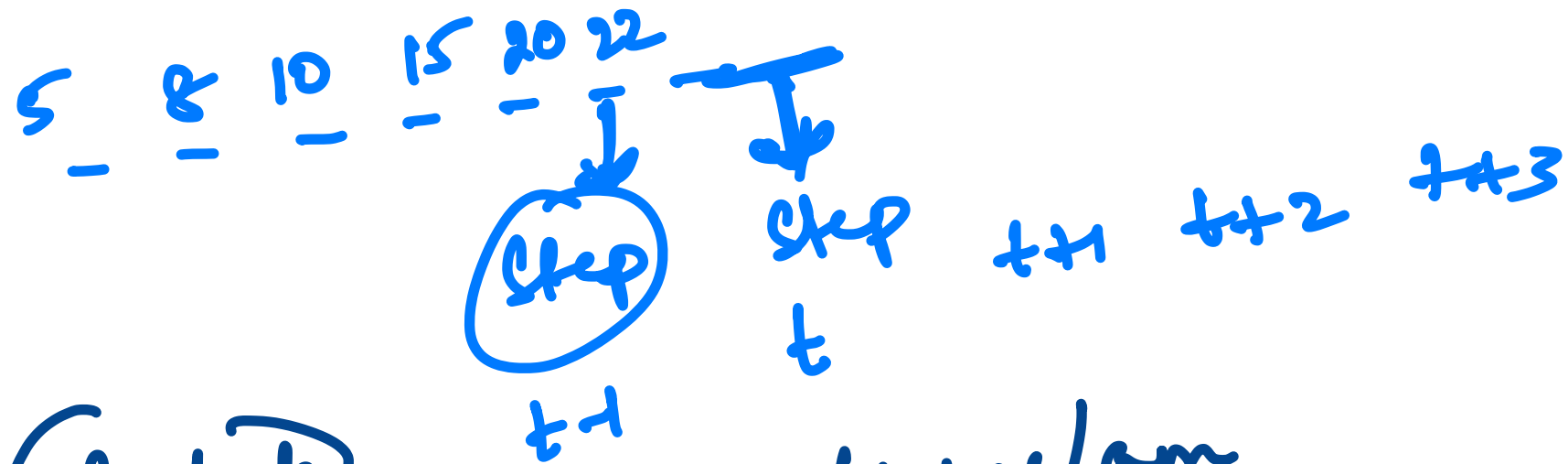
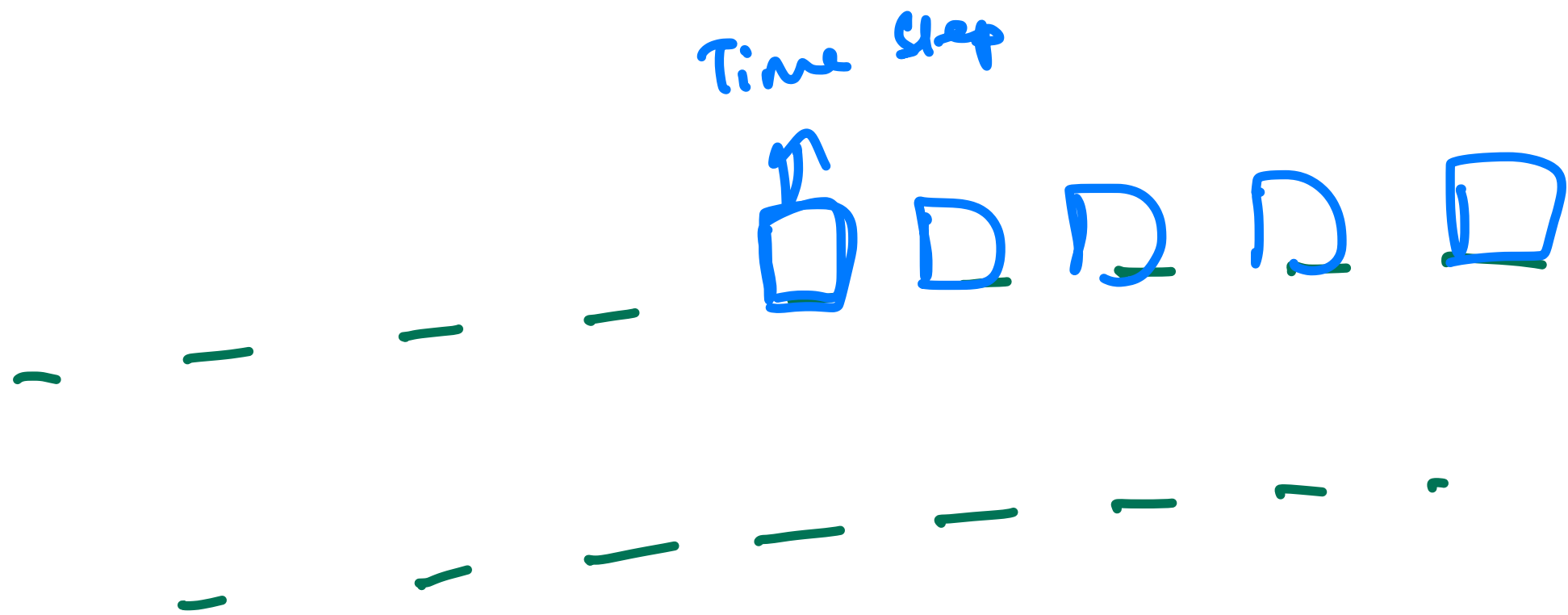
word no. 6

word no. 7

Time Steps

Stock prices / weather

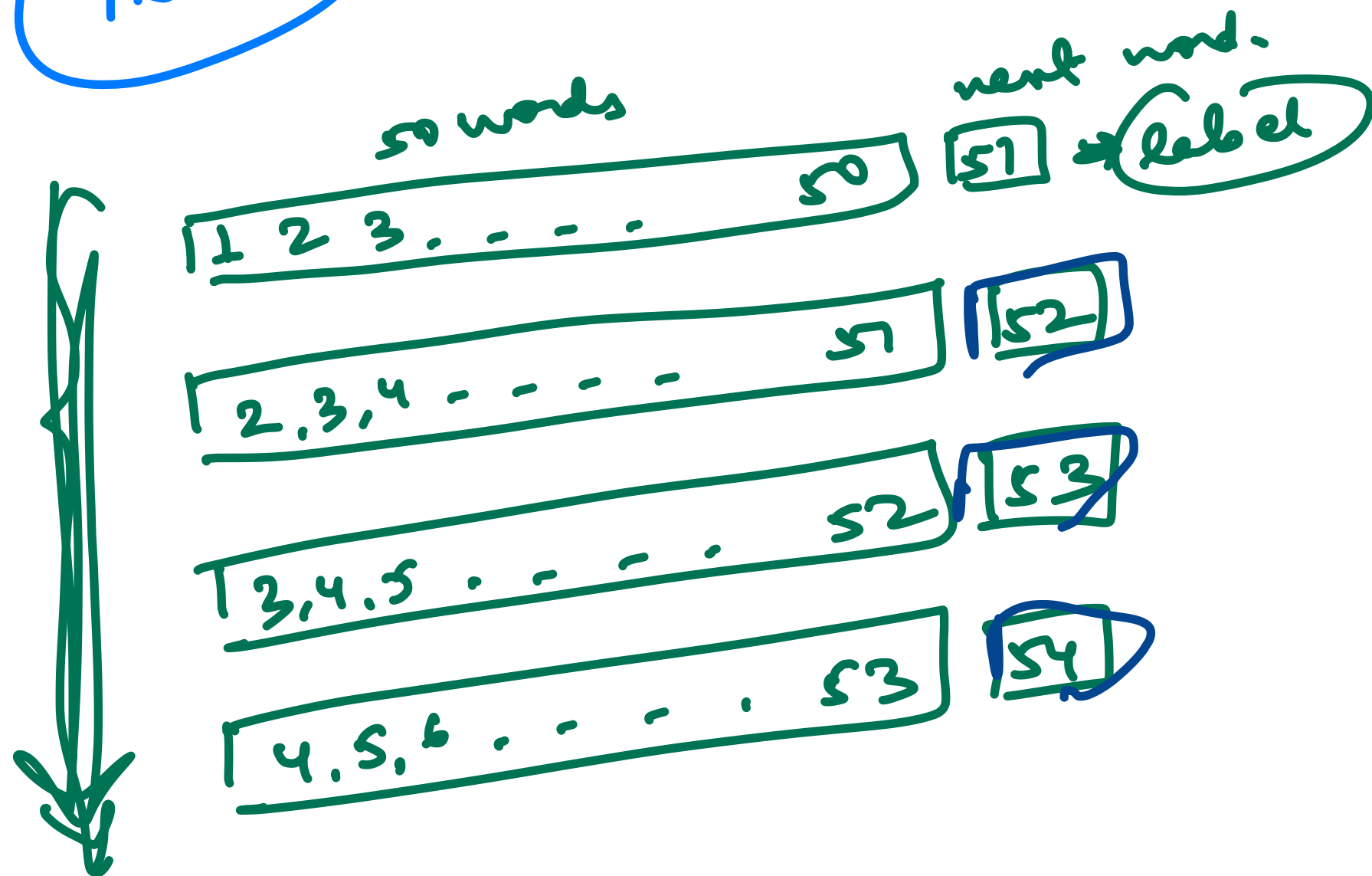
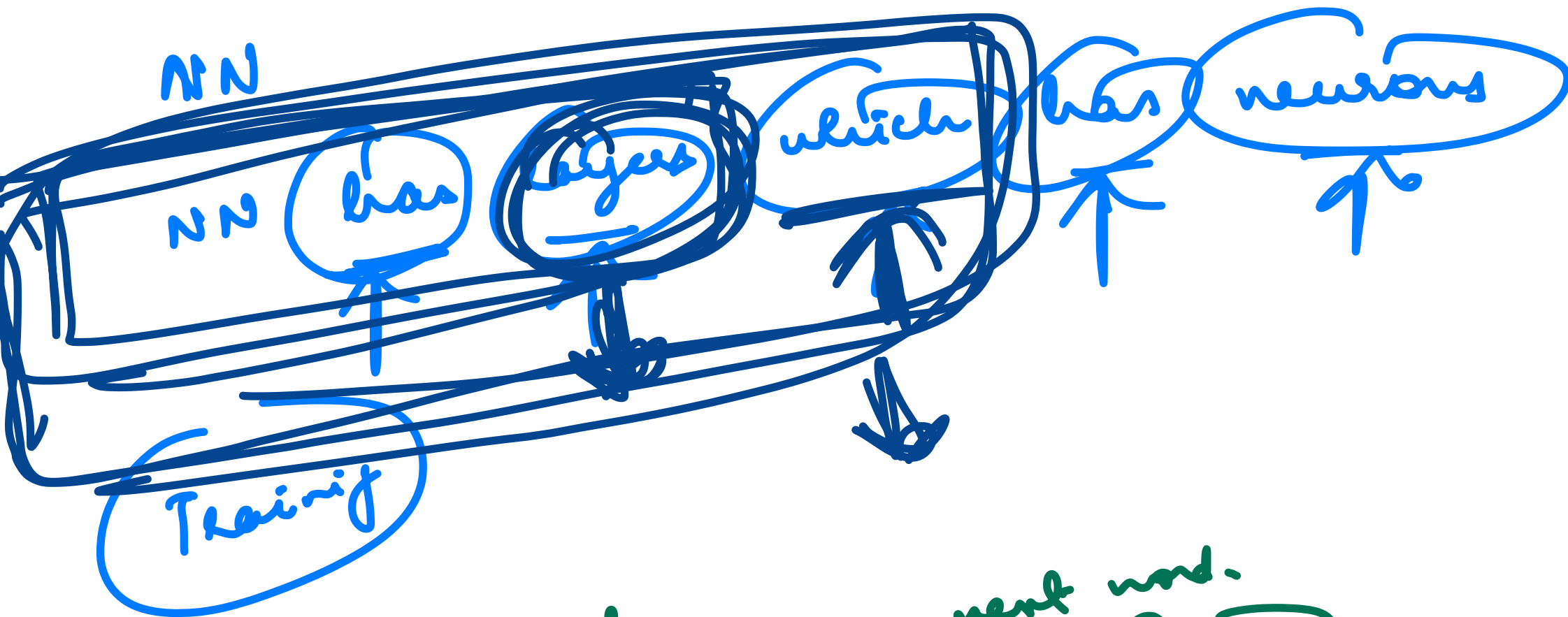
Steps in
Iteration



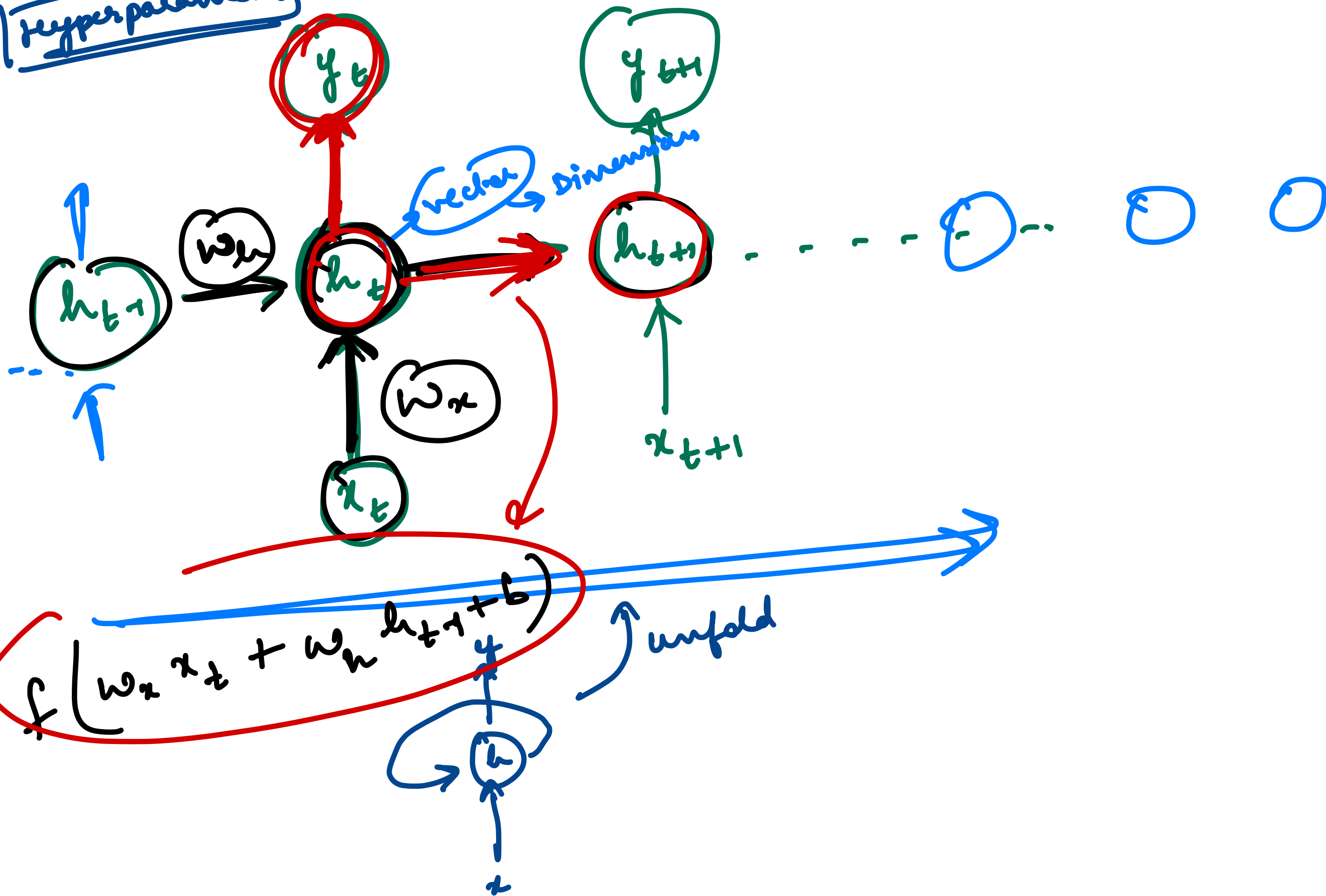
o/p $\rightarrow$ Probable

Context $\rightarrow$ —

hours / eats / likes / am



hyperparameters



$$h_t = f(w_h h_{t-1} + w_x x_t + b)$$

↓
activation function

$$h_t = f(w_h h_{t-1} + w_x x_t + b)$$

hidden state at time step t

↓
Activation func

hidden state at time step $t-1$

wt matrix

wt matrix

ilp at time step t

tanh

↓

o/p $\Rightarrow$ -1 and 1 $\Rightarrow$ normalized values

$$y_t = g(W_y h_t + \underset{\substack{\uparrow \\ \text{bias}}}{c})$$



softmax

//
probabilities for words

output → Simple RNN

vocab → 5 words

I, love, pizza, eat, food

I → [1, 0, 0, 0, 0]

love → [0, 1, 0, 0, 0]

→ Input/
words

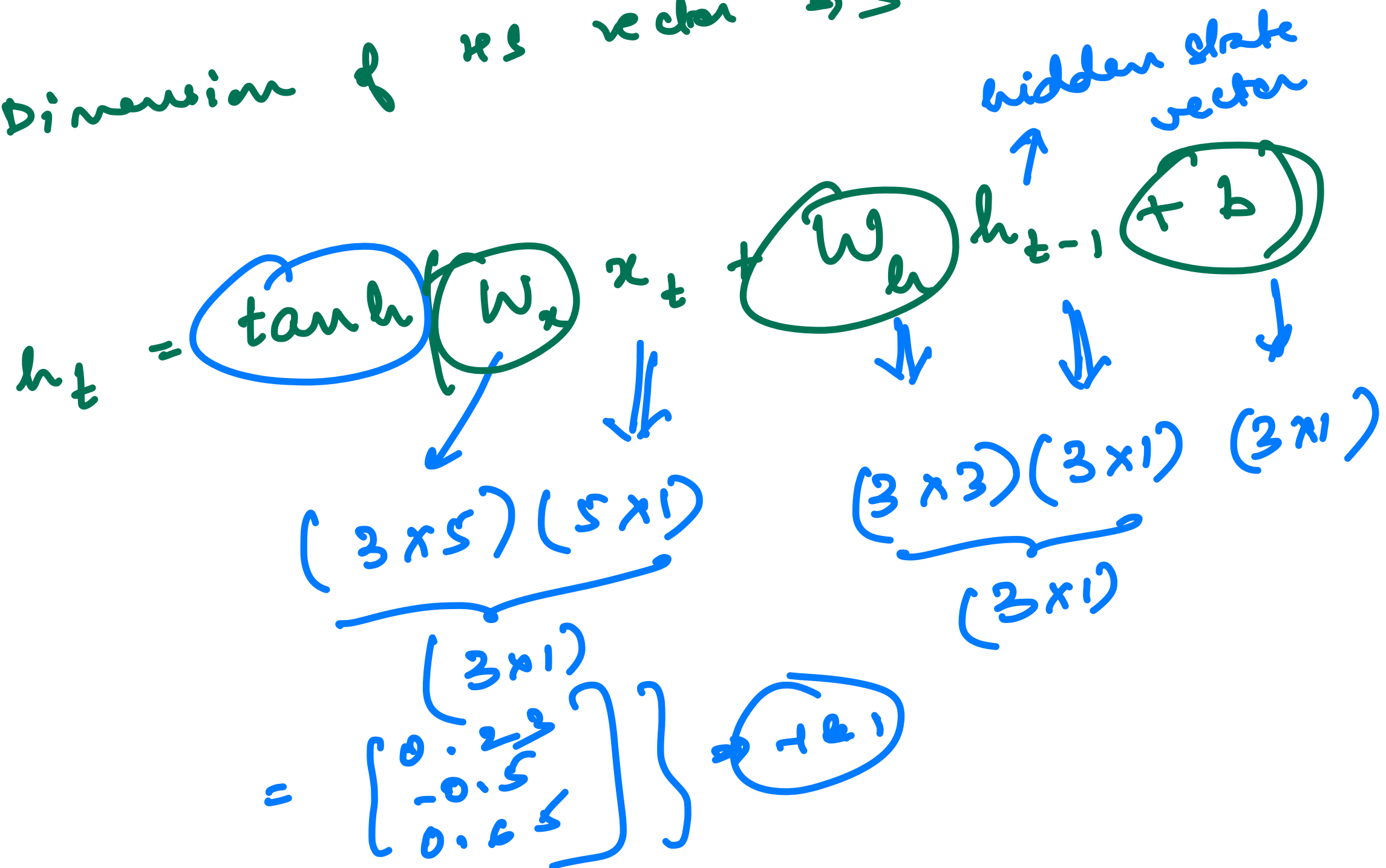
5 Dimensions

give $\rightarrow [0, 0, 1, 0, 0]$

eat $\rightarrow [0, 0, 0, 1, 0]$

food $\rightarrow [0, 0, 0, 0, 1]$

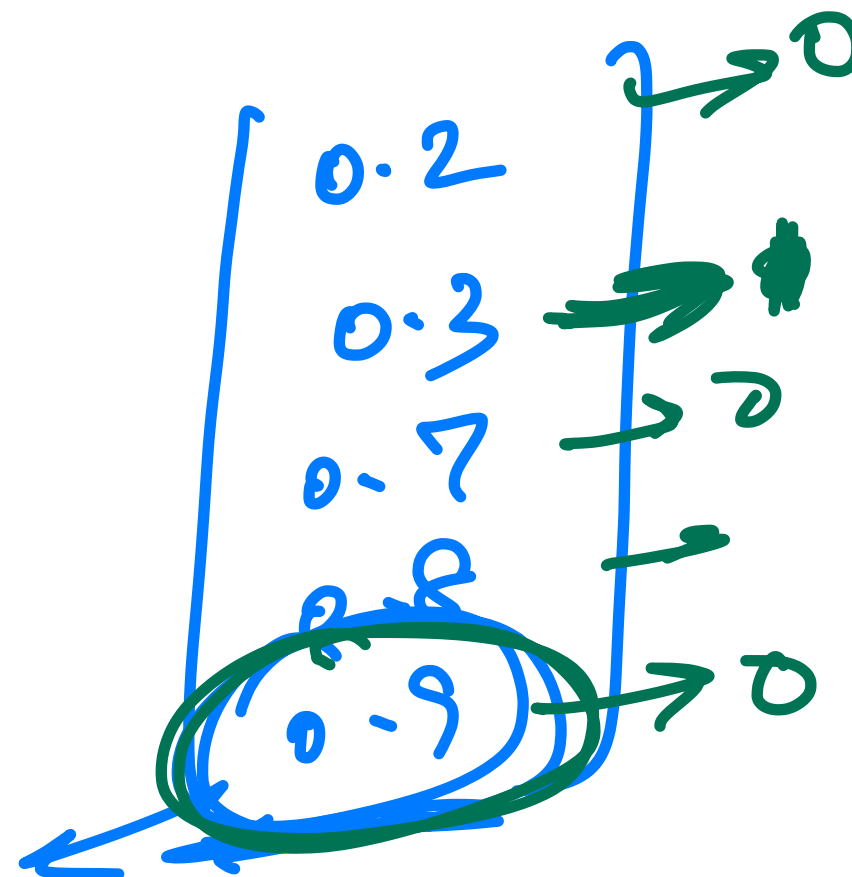
Dimension of x vector $\rightarrow 3$



$$W_x = \begin{bmatrix} 0.2 & 0.3 & 0.1 & 0.7 & 0.4 \\ 0.1 & 0.05 & . & . & . \\ \vdots & . & . & . & . \end{bmatrix}$$

$$y = \text{Softmax}(\underbrace{W}_{(5 \times 3)} h_t + \underbrace{\text{bias}}_{(5 \times 1)})$$

$(5 \times 1) \downarrow$
 Softmax
probab





Keerti

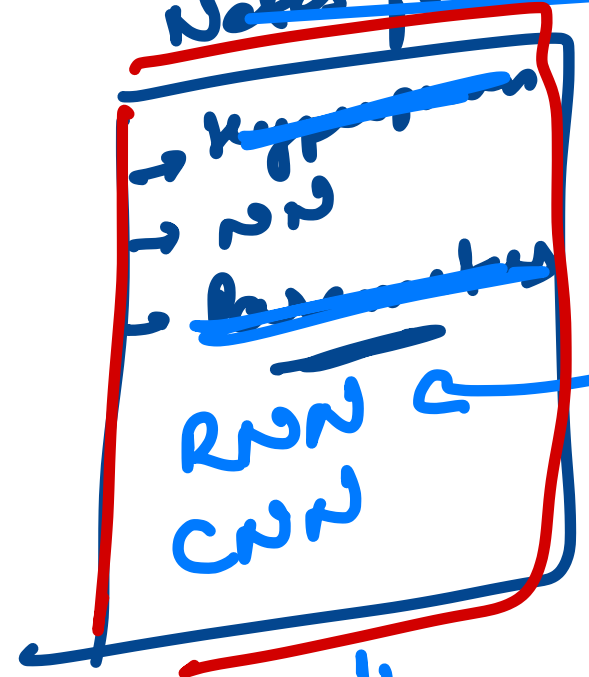
RNN

Short Term Memory
Gajini

Class → Notes for this class

New Input

Notes for Interview Prep



New Notes for Interview Prep

- ① forgetting / removing
- ② Add Input
- ③ calculating off

Novel

10 chapters → 11th chapter

10th chapter

Sathish

↓

brother

age

look

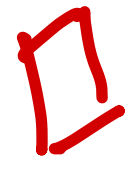
height

forget/remove extra details

Keep relevant → Sathish

Input → Plot for 11th chapter

output → 11th chapter



RNN



forget $\rightarrow$ not imp

keep $\rightarrow$ imp

calculate $\rightarrow$ / 8

LSTM

$\rightarrow$ Special version of RNN

long

Short Term Memory

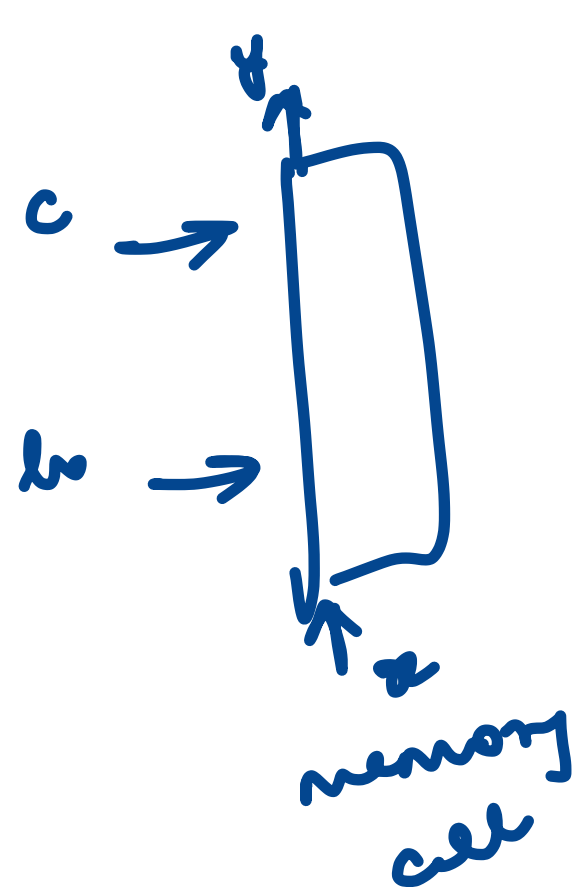
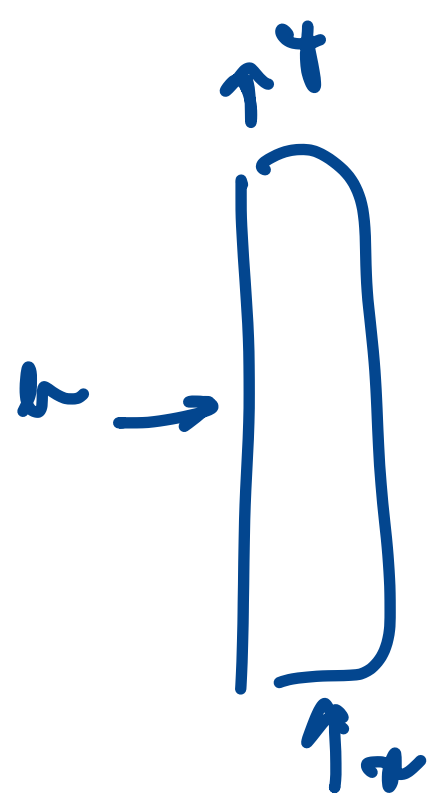
RNN $\rightarrow$ hidden state $\rightarrow$ Short Term Memory

+
Long Term Memory

Cell State

3 Small Networks

- ① Decide what to remove from memory
Forget gate →
 - ② Decide what new info to store
Input gate →
 - ③ Decide what output based on memory
output gate →
- gates → control switches → on & off



① Forget Gate

$$f_t = \sigma(w_f [h_{t-1}, x_t], b_f)$$

0 $\Rightarrow$ forget
1 $\Rightarrow$ keep

② what to store - Input Gate

$$i_t = \sigma(w_i [h_{t-1}, x_t], b_i)$$

new candidate

$$\tilde{c} = (w_c [h_{t-1}, x_t], b_c)$$

③ update cell state

$$c_t = f_t \underline{c_{t-1}} + i_t \tilde{c}_t$$

④ Calculate o/p

$$o_t = \text{Act}(w (h_t, w_o), b)$$

Repeat steps 1-4

RNN
LSTM
GRU

✓ RNN → STM
✓ LSTM → HS
CS ← Complex

⇓
GRU

ChatGPT

BERT

Google

Transformer Architecture

⇓
RNN, LSTM

Theory ←

Work → LLM

App LLM